

Homework 2,3,4



COMS4113

Primary/Backup Key/Value Service

Different / Increased Functionality

(Thanks to Robin Liu (hl3818) , Xijiao Li (xl2950))

Key/value service(hw2, hw3, hw4)

The service maintains a simple database of key/value pairs.
Each key maps to exactly one value.

Three RPCs used to interact with service:

Put, **PutHash**, **Get**

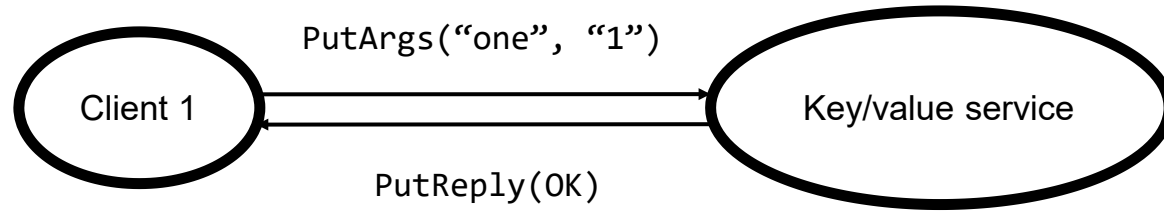
Stateful

Stateful services keep track of data and react differently to the same inputs based on the history.

Key/value service



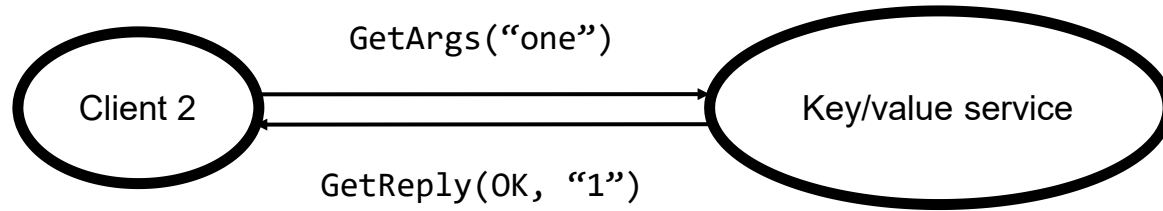
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Key/value service



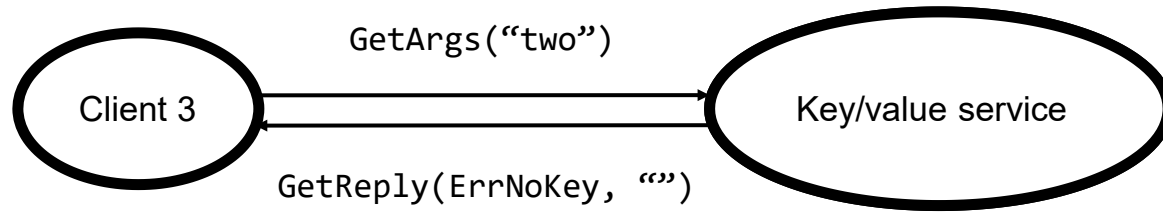
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Key/value service



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HW2: Primary/Backup Replication



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- Motivation: Fault tolerance
- Mechanisms: Backup replication
 - Nominate one server the primary, one server the backup.
If the primary fail, backup will take over
 - Goal #1: Provide a highly reliable service
 - Goal #2: Servers should behave just like a single, more reliable server.
- Need to keep clients, primary, and backup in sync on who is primary and who is backup.

- View service decides who is primary, who is backup.
- A View contains three pieces of info: Viewnum, Primary, Backup
Clients and servers consult view service to find current view.
 - E.g., suppose we have 3 servers labeled S1, S2, and S3. A view could be:
 - (1, S1, S2)
 - (2, S2, S3)
- The viewservice monitors whether each known server is alive or dead.
- For now, assume that view service never fails.

Monitor Server Aliveness



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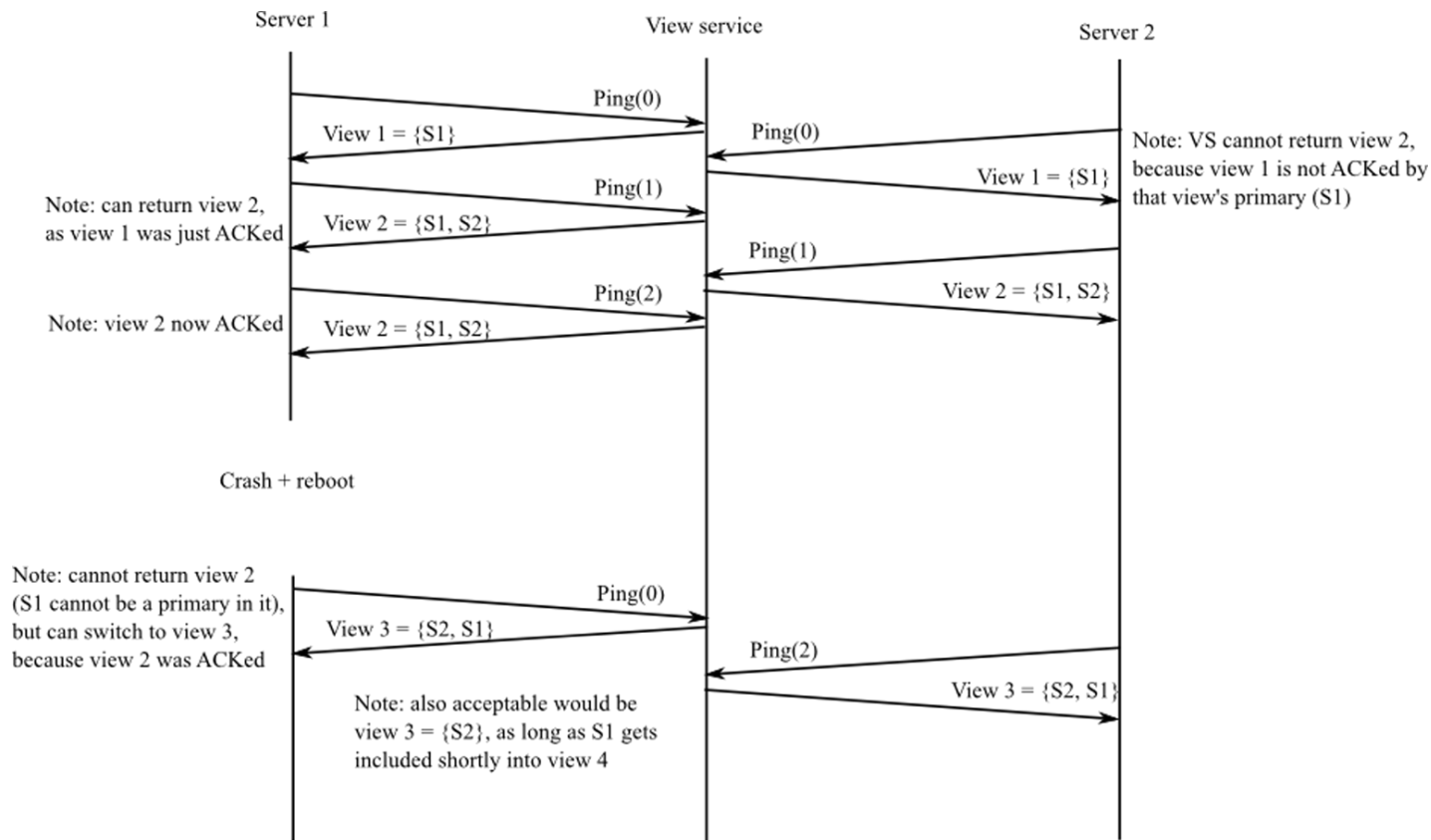
- Each server pings the view server once per **PingInterval**.
 - A **Ping** lets the view server know that the key/value server is alive and the recent **View** it knows about.
 - The view server replies with the current **View**.
- If the view server doesn't receive a **Ping** from a server for a certain number of intervals, it considers the server to be dead.

- The viewservice proceeds to a new view when either
 - at system startup, one server should be promoted to primary (excluded from ACK check)
 - there is no backup and there's an idle server (i.e. a server that sent a **Ping** but is neither the primary nor the backup).
 - it hasn't received a **Ping** from the primary or backup in X PingIntervals;
 - the primary or backup crashed and restarted (**Ping(0)** to viewservice)
- But the view server must **NOT** change views until the primary from the current view acknowledges that it is operating in the current view (i.e. by sending a **Ping** with the current view number).

Example Scenario



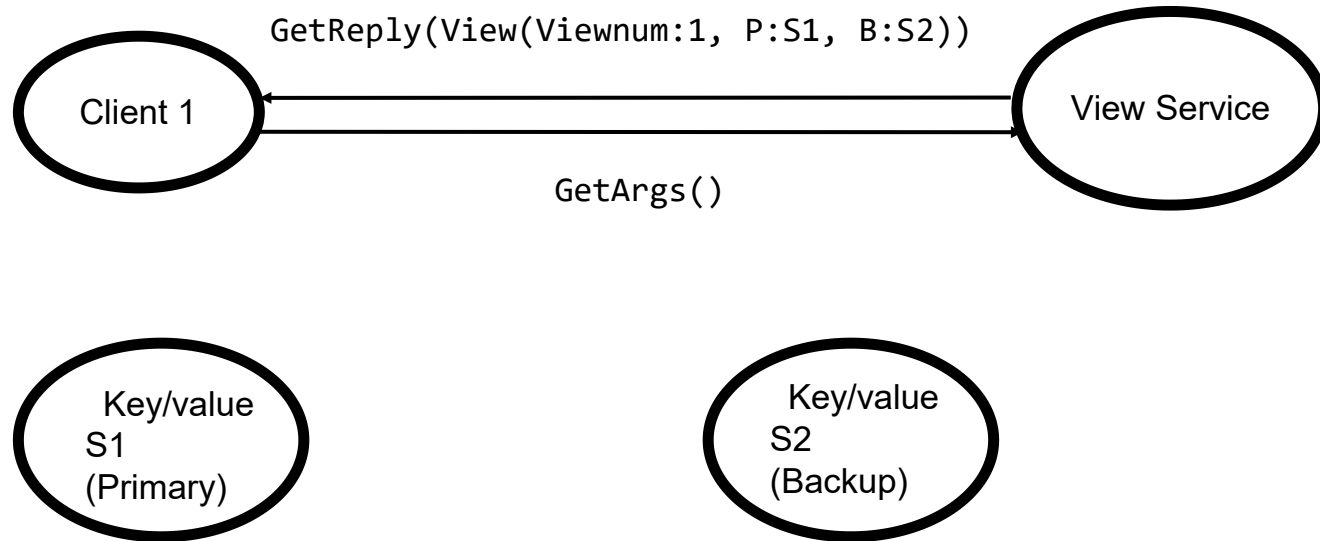
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Part B: Primary/backup Key/value Service



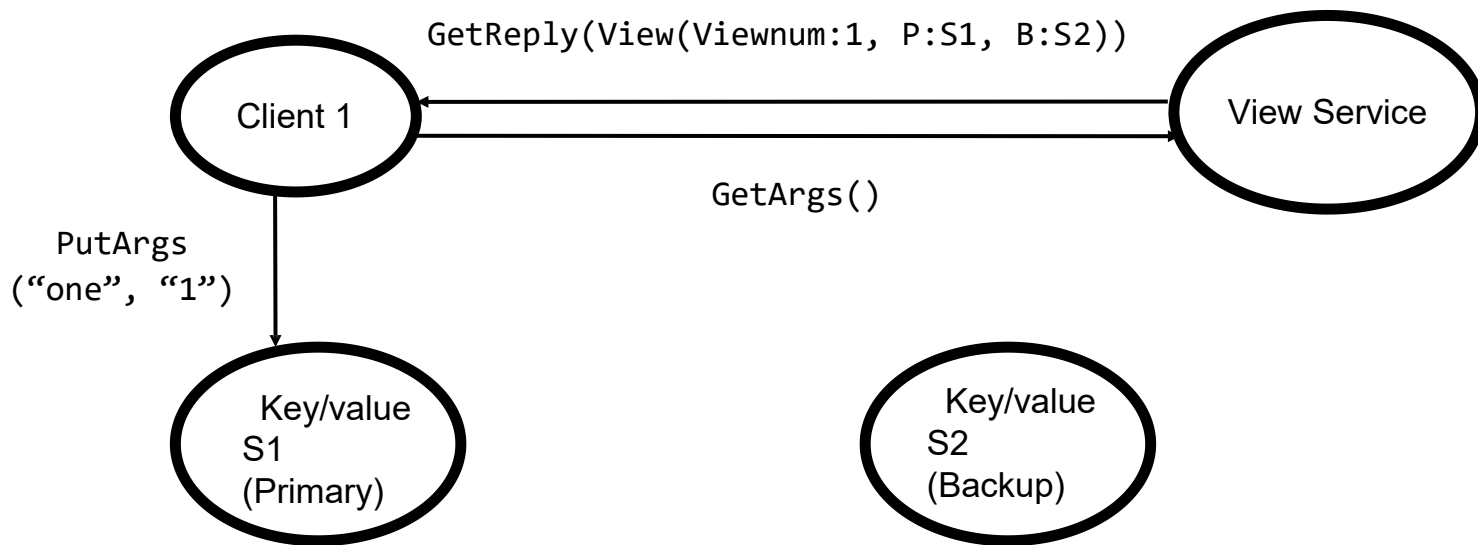
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Part B: Primary/backup Key/value Service



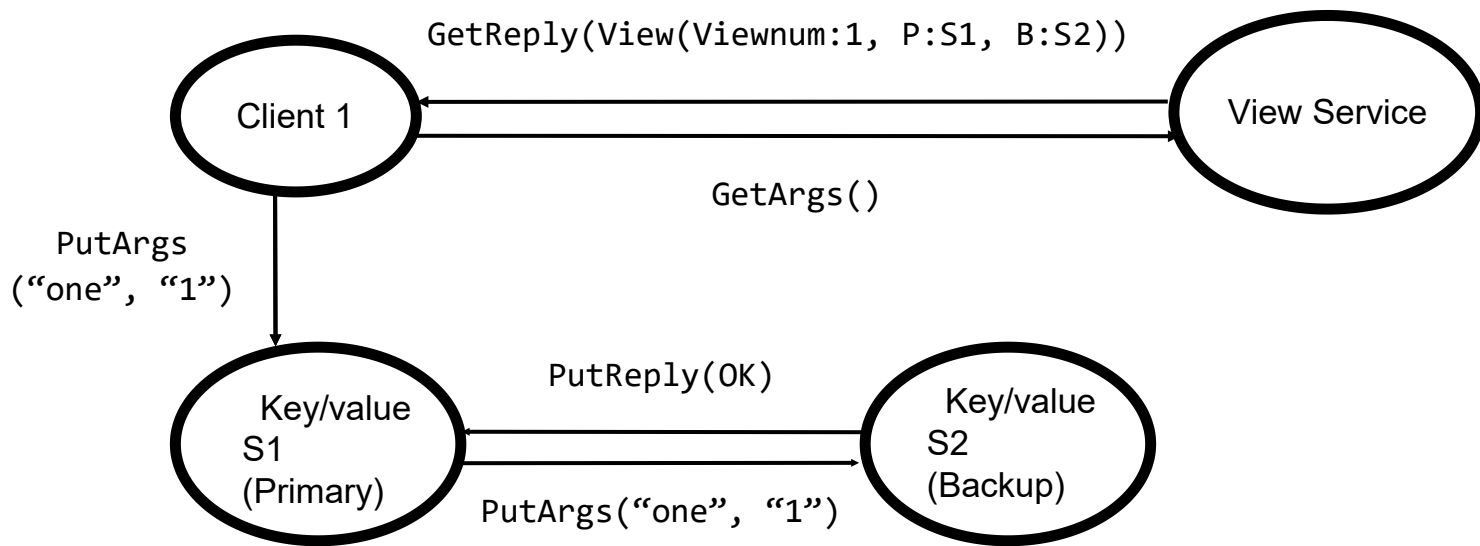
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Part B: Primary/backup Key/value Service



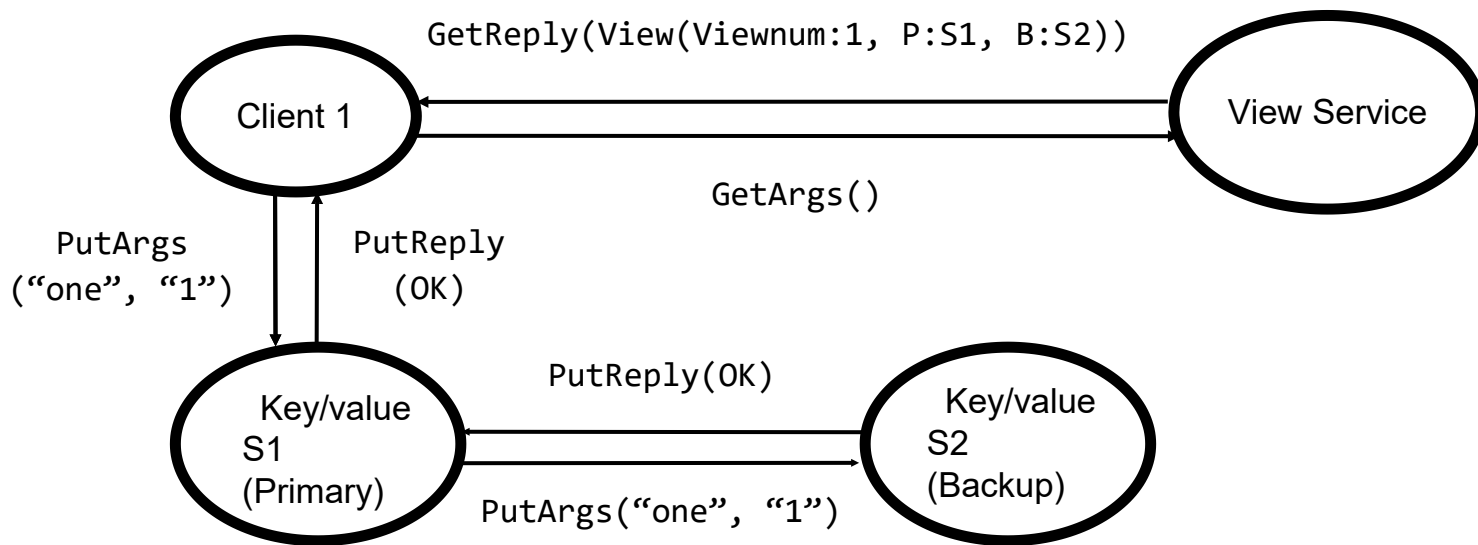
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Part B: Primary/backup Key/value Service



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1. Primary receives the operation from client.
2. Primary sends operation to backup and waits for ACK.
3. Backup performs operation and acks to primary.
4. Primary performs operation and ACK to the client.

Things to think about



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While planning a solution consider:

- Which pieces of data need to be protected from concurrent read/writes (i.e. need a mutex).
- How to detect and filter out duplicate RPC calls from clients.
 - Needed for at-most-once semantic.
- How to safely transfer previous key/value store to server when it becomes primary/backup.

Questions ?



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